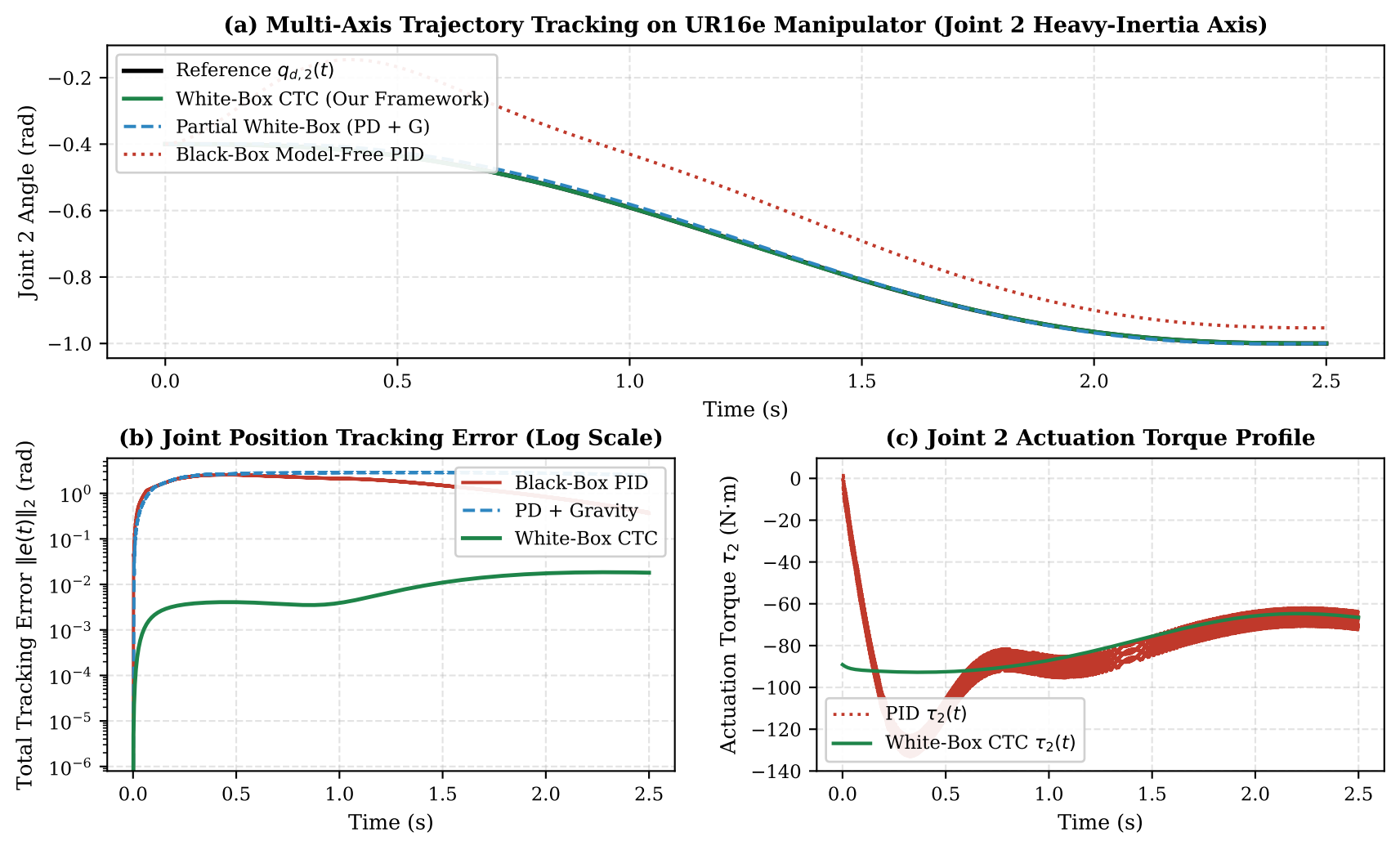

(a) Multi-Axis Trajectory Tracking on UR16e Manipulator (Joint 2 Heavy-Inertia Axis)
Reference $q_{d,2}(t)$
White-Box CTC (Our Framework)
Partial White-Box (PD + G)
Black-Box Model-Free PID
Joint 2 Angle (rad)
−0.2
−0.4
−0.6
−0.8
−1.0
0.0
0.5
1.0
1.5
2.0
2.5
Time (s)
(b) Joint Position Tracking Error (Log Scale)
Black-Box PID
PD + Gravity
White-Box CTC
Total Tracking Error $\|e(t)\|_2$ (rad)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0.0
0.5
1.0
1.5
2.0
2.5
Time (s)
(c) Joint 2 Actuation Torque Profile
PID $\tau_2(t)$
White-Box CTC $\tau_2(t)$
Actuation Torque τ_2 (N·m)
0
−20
−40
−60
−80
−100
−120
−140
0.0
0.5
1.0
1.5
2.0
2.5
Time (s)